
Heart Disease Prediction System

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1. Problem Description

Heart disease is one of the leading causes of death worldwide. Early detection and prediction of heart disease can help doctors take preventive measures and provide timely treatment to patients. However, analyzing large amounts of medical data manually can be difficult and time-consuming. Machine Learning techniques can help in identifying patterns in medical data and predicting whether a person is at risk of heart disease. The aim of this project is to design and develop an end-to-end Machine Learning classification system that predicts the presence of heart disease using clinical patient data. The system will use a heart disease dataset containing medical attributes such as age, blood pressure, cholesterol level, chest pain type, and heart rate. The dataset will be preprocessed by handling missing values, normalizing numerical data, and converting categorical variables into numerical form. Different Machine Learning models such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) will be trained on the dataset. The performance of these models will be evaluated using metrics like accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. The final model will help in predicting whether a patient has heart disease (1) or not (0).

2. Architecture

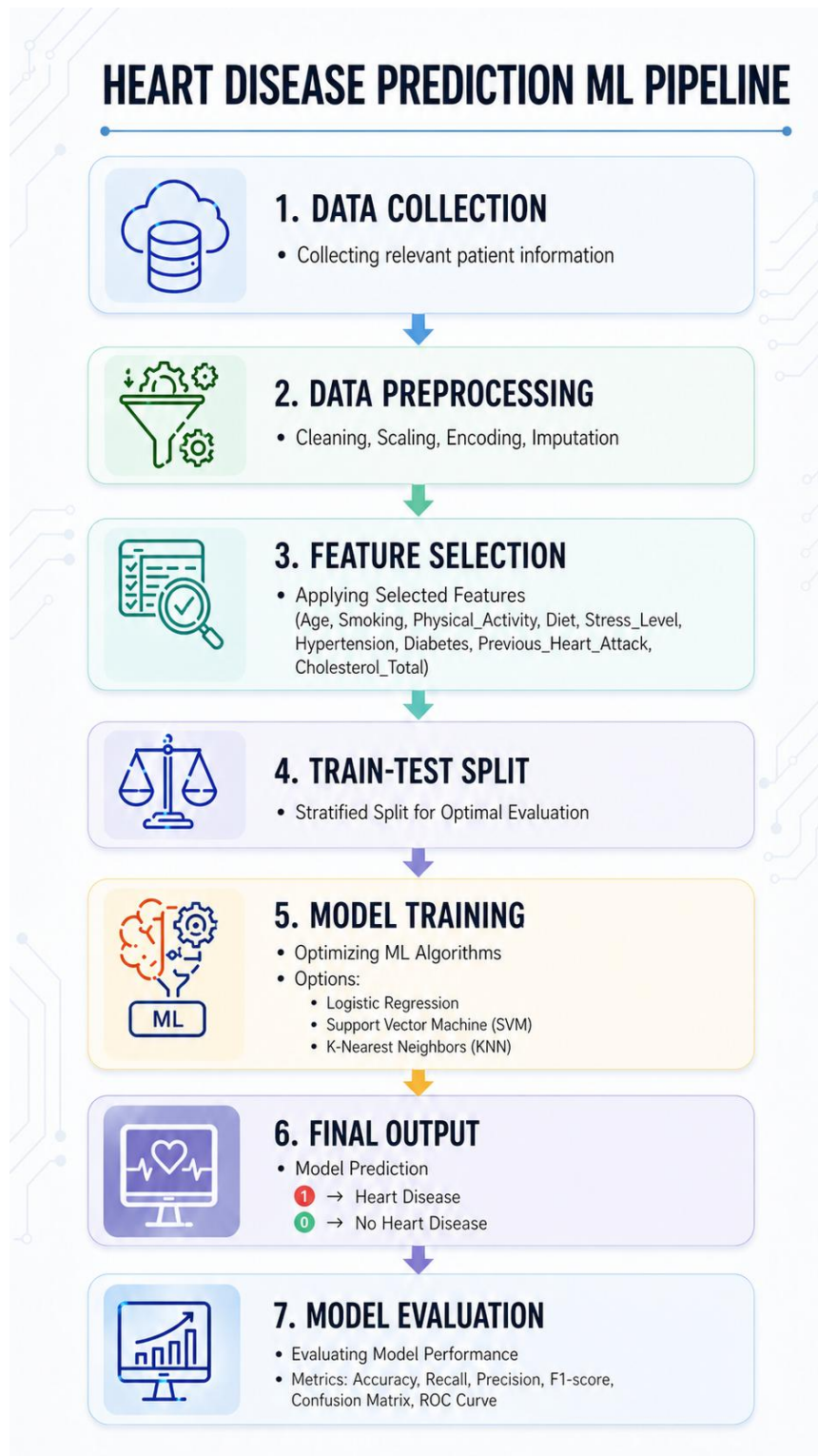


Fig. 1 Architecture for Heart Disease Prediction

3. Models Used:

- a) **Logistic Regression:** Logistic Regression is a powerful statistical algorithm widely utilized for classification tasks. It works by applying a logistic (sigmoid) function to the input features to estimate the probability of a data point belonging to a specific category.
- b) **K-Nearest Neighbors (KNN):** KNN is an intuitive, non-parametric, and instance-based learning algorithm. Rather than building a rigid mathematical equation during training, it stores the dataset and makes predictions by calculating the distance (usually Euclidean) between a new data point and all existing points.
- c) **Support Vector Machine (SVM):** SVM is a robust algorithmic approach that maps data points into a high-dimensional space to discover the optimal "hyperplane"—the mathematical boundary that provides the widest possible gap between different classes.

4. Evaluation metrics used:

- a) **Accuracy:** This is the foundational metric of our evaluation. It calculates the overall correctness of the model by dividing the number of correct predictions (both True Positives and True Negatives) by the total number of predictions made.
- b) **Precision:** Precision measures the accuracy of positive predictions. High precision indicates a low false-positive rate.
- c) **Recall (Sensitivity):** Recall measures the model's ability to find all relevant instances within a dataset.
- d) **F1-Score:** The F1-Score is the harmonic mean of Precision and Recall. We consider this metric because it provides a single, balanced score that accounts for both false positives and false negatives.
- e) **Confusion Matrix:** While not a single mathematical score, the Confusion Matrix is a critical visual evaluation tool we utilized. It provides a detailed table showing exactly where a model is making errors by mapping the predicted classes against the actual classes.
- f) **ROC Curve (Receiver Operating Characteristic):** The ROC curve is a graphical representation of a model's performance across all possible classification thresholds. It plots the True Positive Rate (how many actual positives were correctly identified) against the False Positive Rate (how many negatives were incorrectly labeled as positive).
- g) **AUC (Area Under the Curve):** AUC is the single numerical value that summarizes the ROC curve. It measures the entire two-dimensional area underneath the entire ROC curve, providing an aggregate measure of performance. The score ranges from 0 to 1, where 1.0 represents a perfect model that makes zero mistakes.

5. Preliminary Results and Visualizations

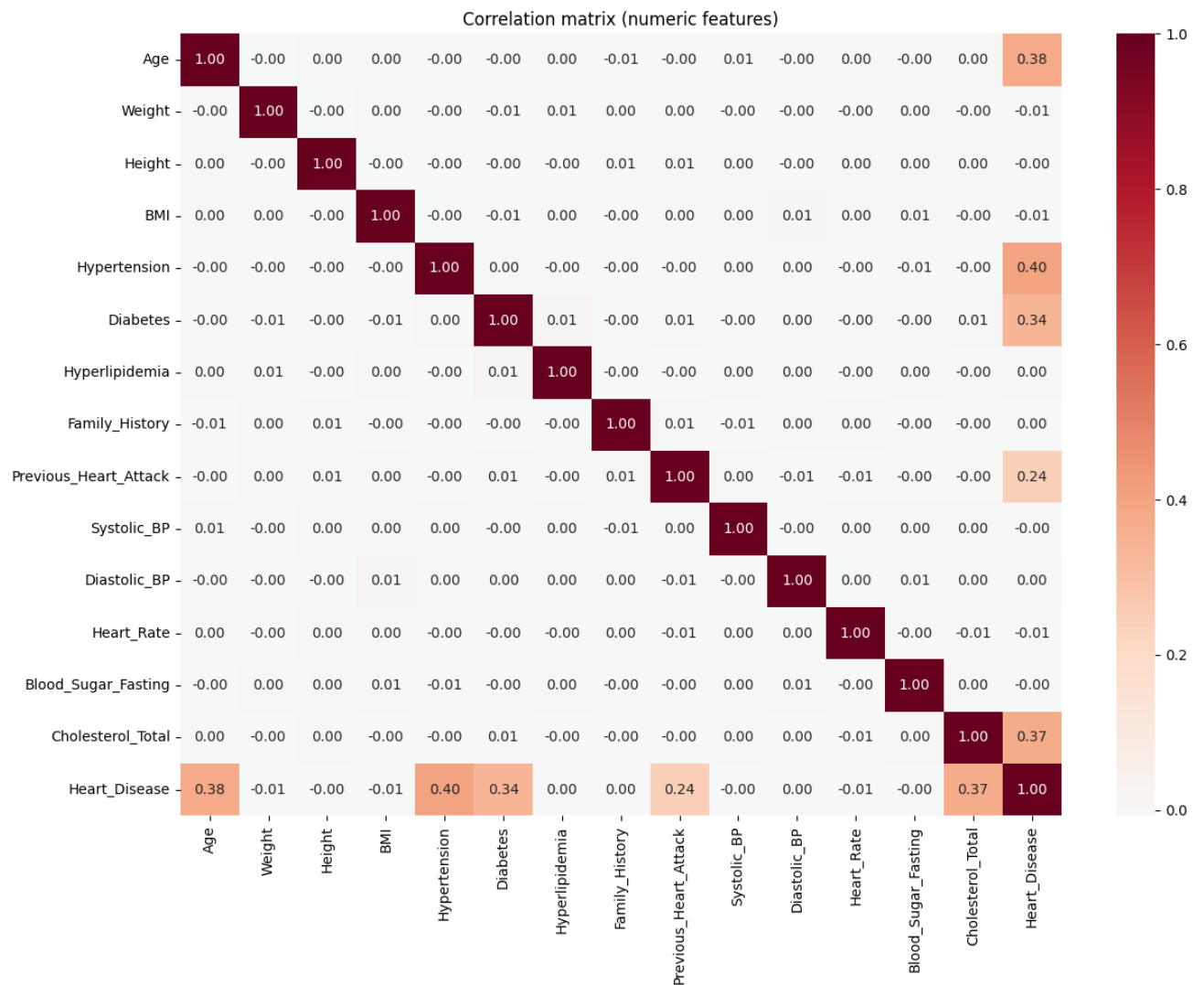


Fig.2 Heatmap showing correlation.

Classification Report:					
	precision	recall	f1-score	support	
0	0.93	0.93	0.93	5365	
1	0.92	0.92	0.92	4635	
accuracy			0.92	10000	
macro avg	0.92	0.92	0.92	10000	
weighted avg	0.92	0.92	0.92	10000	

Fig.3 Logistic Regression

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	precision	recall	f1-score	support	
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Fig.4 KNN

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weighted avg	0.93	0.93	0.93	10000	

Fig.5 SVM

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>